

Customer Purchase Behavior Study

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1. Project Overview

This study examines consumer purchasing patterns using transactional records from 3,900 shopping entries spread across multiple product categories. The primary objective is to extract meaningful insights related to expenditure trends, customer groupings, product demand, and membership behavior — all of which can support data-driven strategic planning for retail businesses.

2. Dataset Summary

The dataset used in this analysis consists of the following attributes:

- **Total Records:** 3,900
- **Total Attributes:** 18
- **Key Data Fields**
 - Demographic information: Age, Gender, Location, Subscription Status
 - Transaction details: Item Purchased, Category, Purchase Amount (USD), Season, Size, Color
 - Behavioral attributes: Discount Applied, Promo Code Usage, Previous Purchases, Purchase Frequency, Review Rating, Shipping Preference
- **Data Quality Issue:** 37 entries had missing values in the Review Rating field

3. Exploratory Data Analysis using Python

The data preparation and exploration phase was carried out entirely in Python, following the steps outlined below:

- **Data Import:** Loaded the dataset into a pandas DataFrame for processing.
- **Preliminary Inspection:** Utilized
 - `df.info()` to examine the data types and structure
 - `.describe()` to obtain basic statistical measures
- **Handling Missing Values:** Identified null entries and filled them in the Review Rating column by substituting the median rating calculated per product category.

- **Column Renaming:** Standardized all column headers to snake_case format to improve readability and consistency.
- **Feature Engineering**
 - **age_group:** Derived by segmenting customer ages into defined bins
 - **purchase_frequency_days:** Calculated from available purchase frequency data
- **Redundancy Check:** Examined whether discount_applied and promo_code_used carried duplicate information; removed promo_code_used as it was found to be redundant.
- **Database Export:** Established a connection between Python and PostgreSQL, then pushed the preprocessed DataFrame into the database to enable SQL-based querying.

4. Data Analysis using SQL (Business Transactions)

Structured queries were executed in PostgreSQL to address the following key business questions:

1. **Revenue by Gender** – Evaluated and compared total sales revenue generated by male versus female customers.

gender (text)	revenue (numeric)
Female	75191
Male	157890

2. **High-Spending Discount Users** – Pinpointed customers who applied discounts yet had purchase amounts exceeding the overall average.

customer_id (bigint)	purchase_amount (bigint)
2	64
3	73
4	90
7	85
9	97
12	68

13	72
16	81
20	90
22	62
24	69

Total rows: 839 | Query completed successfully

- Top 5 Products by Rating** – Determined which products received the highest average review scores from customers.

item_purchased (text)	Average Product Rating (numeric)
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.80
Skirt	3.78

- Shipping Type Comparison** – Assessed differences in average spending between customers who chose Standard versus Express delivery.

shipping_type (text)	avg_purchase (numeric)
Standard	58.46
Express	60.48

- Subscribers vs. Non-Subscribers** – Contrasted average spending and overall revenue across customers with and without active subscriptions.

subscription_status	total_customers	avg_spend	total_revenue
Yes	1053	59.49	62645.00
No	2847	59.87	170436.00

- Discount-Reliant Products** – Highlighted the 5 products with the greatest proportion of purchases made using a discount.

item_purchased (text)	discount_rate (numeric)
Hat	50.00
Sneakers	49.66
Coat	49.07
Sweater	48.17
Pants	47.37

7. **Customer Segmentation** – Grouped customers into New, Returning, and Loyal categories based on their transaction history.

customer_segment (text)	Number of Customers (bigint)
Loyal	3116
New	83
Returning	701

8. **Top 3 Products per Category** – Ranked the most frequently purchased items within each product category.

item_rank	category	item_purchased	total_orders
1	Accessories	Jewelry	171
2	Accessories	Sunglasses	161
3	Accessories	Belt	161
1	Clothing	Blouse	171
2	Clothing	Pants	171
3	Clothing	Shirt	169
1	Footwear	Sandals	160
2	Footwear	Shoes	150
3	Footwear	Sneakers	145
1	Outerwear	Jacket	163
2	Outerwear	Coat	161

9. **Repeat Buyers & Subscription Likelihood** – Investigated whether customers with more than 5 prior purchases tend to have active subscription status.

subscription_status (text)	repeat_buyers (bigint)
No	2518
Yes	958

10. **Revenue by Age Group** – Computed the total revenue contribution attributed to each customer age bracket.

age_group (text)	total_revenue (numeric)
Young Adult	62143
Middle-aged	59197
Adult	55978
Senior	55763

5. Dashboard in Power BI

An interactive dashboard was developed in Power BI to visually communicate the analytical findings. The dashboard includes slicers for Subscription Status, Gender, Category, and Shipping Type, along with KPI cards and charts covering revenue by category, age group, and gender.

Key metrics displayed in the dashboard:

- **Number of Customers:** 3.9K
- **Average Review Rating:** 3.75
- **Average Purchase Amount:** \$59.76
- **Subscription split:** 73% Non-subscribers, 27% Subscribers
- **Top revenue category:** Clothing (\$0.10M)



6. Business Recommendations

- **Grow the Subscriber Base:** Design and promote exclusive perks and loyalty rewards targeted at non-subscribers to increase membership sign-ups.
- **Strengthen Customer Retention:** Introduce loyalty programs to transition Returning customers into the Loyal segment through incentivized repeat purchases.
- **Revisit Discount Strategy:** Carefully evaluate the discount policy to ensure promotional campaigns drive revenue without significantly eroding profit margins.
- **Leverage Top-Performing Products:** Feature highest-rated and best-selling items more prominently in marketing campaigns and product listings.
- **Precision-Targeted Campaigns:** Allocate marketing efforts toward high-revenue age groups (Young Adults and Middle-aged) and customers who prefer Express shipping, as these segments demonstrate stronger spending behavior.